

A Four-Element High Isolation UWB MIMO Antenna for Sub-6 GHz, WiMAX, and WLAN Applications

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Abstract— A four-element high-isolation Ultra-Wideband (UWB) MIMO antenna is reported for Sub-6 GHz, WiMAX, and WLAN systems. The MIMO antenna consists of rectangular, slotted, irregular octagonal radiating patches with a partial ground plane. It is designed on a Rogers RT5880 (lossy) with a size of $80 \times 80 \times 0.79$ mm³. The irregular octagon-shaped antenna shows a resonant response at 3.16 GHz and 5.58 GHz, achieving excellent reflection coefficients of -32 dB and -44.4 dB, respectively, along with a wide bandwidth of 7.32 GHz from 2.72 to 10.04 GHz. It exhibits a gain of 4.87 dB (Max.) and an efficiency of 94% (Max.). In addition, the four-element MIMO antenna provides good isolation between the radiating elements, with the lowest isolation of -19.5 dB and the highest isolation of -32 dB. On account of its compact structure, high isolation, wide bandwidth, and optimized radiation performance, the designed MIMO antenna is suitable for the intended multi-band applications.

Keywords—UWB, MIMO, High isolation, Sub-6GHz, WLAN, WiMAX.

I. INTRODUCTION

The development of communication systems has played a vital role in the progress of human civilization, particularly from the nineteenth century onward. The introduction of electrical circuits and telegraph systems significantly transformed long-distance communication by improving its speed, reliability, and efficiency [1]. Later, pioneering contributions to wireless communication, including the experimental work of J. C. Bose and G. Marconi, demonstrated the practical potential of transmitting electromagnetic waves without physical conductors [2]. Since then, wireless communication technology has continued to evolve rapidly to address the increasing demand for high-speed, reliable, and compact systems. An antenna is a fundamental component of any RF system, as it enables the propagation and reception of radio signals through free space. To achieve higher data rates, improved link reliability, and enhanced channel capacity, modern wireless systems

increasingly employ MIMO technology, where multiple antenna elements are used at the transmitting and/or receiving ends [3-4]. MIMO technology is especially important for advanced wireless standards such as WLAN, WiMAX, and 5G, where compact size, wide bandwidth, good radiation performance, and high data throughput are essential design requirements.

In an ideal propagation environment, increasing the number of MIMO elements can improve system capacity and diversity performance. However, in practical compact devices, closely spaced antenna elements often suffer from strong mutual coupling. This coupling can degrade impedance matching, efficiency, ECC, diversity gain, and overall MIMO performance [5-6]. Therefore, achieving high isolation among closely packed antenna elements while maintaining compact size, wide bandwidth, and stable radiation characteristics remains a major challenge in MIMO system [7-8].

Several studies [9]-[16] have addressed mutual coupling and performance limitations in MIMO antenna design by proposing different configurations with trade-offs among antenna size, operating bandwidth, gain, efficiency, isolation, and structural complexity. For instance, T. Addepalli et al. presented a relatively large MIMO configuration operating in the lower 5G band. Although the antenna achieves excellent efficiency and acceptable isolation between the radiating elements, it exhibits a comparatively low gain of more than 2.5 dB and covers only the lower 5G frequency region [9]. In [10], S. H. Kiani et al. developed a MIMO system with a large footprint of $150 \times 75 \times 0.8$ mm³; however, the antenna shows relatively low gain of 3.5-3.9 dB and moderate efficiency of 58-72%. A MIMO antenna with good inter-element isolation of -18.8 dB was demonstrated in [11]. Nevertheless, despite its reasonable gain, the antenna has a large size of $120 \times 65 \times 1.6$ mm³ and covers only a limited bandwidth within a portion of the lower 5G band, which limits its suitability for current wideband wireless requirements. S. Padmanathan et al. introduced a large two-element MIMO antenna covering a wide bandwidth of 5.2 GHz with acceptable isolation;

however, the antenna exhibits low gain of 2.97 dB and low efficiency for Sub-6 GHz applications [12]. Another MIMO antenna proposed in [13] was designed for 5G smartphone with a large size of $150 \times 75 \times 1.524$ mm³. Although the design improves radiation performance and provides high isolation with an efficiency of 90–92%, its operating band covers only a limited portion of the lower 5G region. In addition, the comparatively low gain reported in [13]–[15] remains a limitation. A 10-element MIMO system operating over a narrow range of 3.4–3.6 GHz was presented in [15], but its gain remains within 2–4 dB and the efficiency is comparatively low. In [16], U. Rafique et al. proposed a MIMO configuration with good efficiency of around 90%; however, the antenna provides a limited gain of 3.64 dB and relatively poor isolation within a size of $150 \times 75 \times 1.6$ mm³.

In this study, a compact four-element high-isolation ultra-wideband MIMO configuration is presented for Sub-6 GHz, WiMAX, and WLAN applications. This multi-element antenna significantly improves channel capacity and diversity gain without making the system costly. Using more than four antennas gives diminishing returns, while four-element MIMO (e.g., 4×4) is practical for achieving high data rates and reliable signal transmission [17]. This design developed with a simple structure and low design complexity while maintaining a wide operating bandwidth within the targeted application bands. The four-element MIMO antenna has a size of 80×80 mm² and is made on a Rogers RT5880 with a relative permittivity of 2.2 and a thickness of 0.79 mm. The radiating elements achieve good isolation, with a min. isolation of -19.5 dB and a max. isolation of -32 dB. In addition, the antenna provides a maximum gain of 4.87 dB along with high efficiency, confirming its suitability for compact wideband wireless systems.

II. ANTENNA DESIGN METHODOLOGY

A. Single Element Configuration

Fig. 1 illustrates the topology of the single element. The antenna consists of 3 main parts: the radiating patch, substrate, and ground plane. The single-element antenna has an overall size of 40×40 mm². The antenna is excited by a rectangular feed line with a characteristic impedance of 50 Ω . The radiating element is initially designed as a 24×24 mm² square-shaped copper patch on the top side of the substrate. To obtain the optimized geometry, portions of metal are removed from all four edges of the square patch, forming an irregular octagonal-shaped radiating patch, as shown in Fig. 1. A rectangular slot of 14×10 mm² is then introduced inside the octagonal patch to facilitate the impedance matching and BW performance. In addition, a partial metallic ground plane is used on the underside of the substrate to enhance the reflection coefficient characteristics. In Fig. 2(a), the optimized single antenna element operates over a wide range from 2.72 GHz to 10.15 GHz. The antenna exhibits reflection coefficients of -26.92 dB at 3.21 GHz and -43.675 dB at 5.676 GHz, indicating excellent impedance matching at both resonant frequencies. Fig. 2(b) examines the effect of the width (G_1) on the $|S_{11}|$ response. It is observed that the S-parameter characteristics vary with the value of G_1 , and the best performance is achieved when $G_1 = 14.50$ mm, which is selected as the final value for the proposed design. The optimized geometrical parameters of the proposed antenna are presented in Table I.

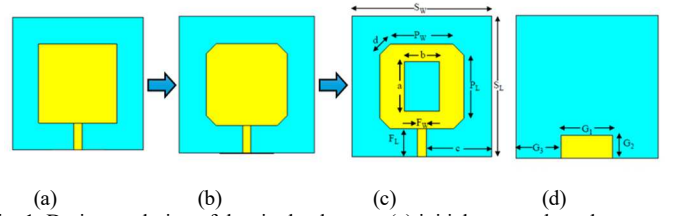


Fig. 1. Design evolution of the single-element: (a) initial square-shaped patch, (b) modified patch with chamfered edges, (c) slotted optimized top-view geometry of the element, and (d) corresponding ground-plane configuration.

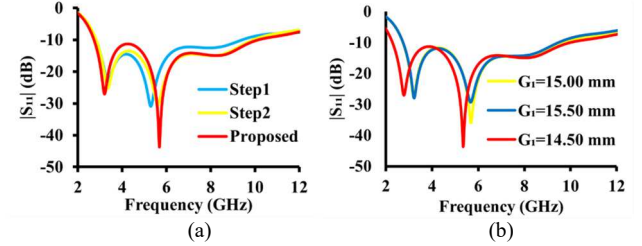


Fig. 2. Reflection coefficient analysis of the element: (a) $|S_{11}|$ comparison for the three design steps and (b) $|S_{11}|$ variation with ground-plane width G_1 .

TABLE I. OPTIMIZED DIMENSIONS OF THE SINGLE ELEMENT DESIGN

Symbol	Dimension (mm)	Symbol	Dimension (mm)
$S_W = S_L$	40	b	10
P_W	18	c	18.77
P_L	18	d	4.24
F_L	08	G_1	14.50
F_W	2.46	G_2	6.50
a	14	G_3	12.75

B. 4-Element MIMO Configuration

A MIMO configuration comprising four elements has been constructed, with its design derived from a single antenna as a reference. According to the investigation in [17], a 4×4 MIMO configuration offers an effective trade-off between transmission reliability and hardware complexity. Increasing the number of antennas beyond four provides only marginal performance gains. Fig. 3 illustrates the configuration of the MIMO antenna. To attain enhanced isolation, the placement of identical antennas is arranged in an orthogonal manner. The dimensions of MIMO antennas are 80×80 mm². Four individual ports are used for providing excitation to the four elements antenna using rectangular line feed with an impedance of 50 Ohm. Table II shows required dimensions of the MIMO antenna.

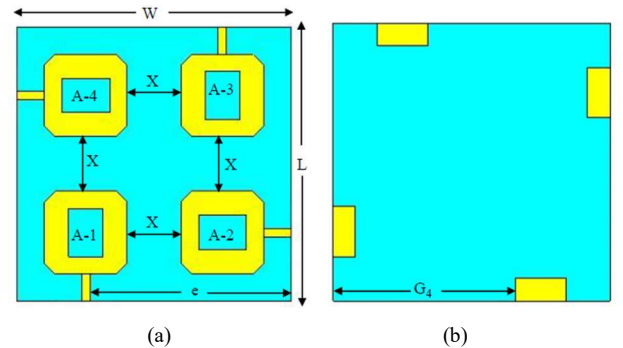


Fig. 3. Layout of the MIMO configuration. (a) Front and (b) Back view.

TABLE II. DIMENSIONS OF THE MIMO DESIGN

Symbol	Dimension (mm)	Symbol	Dimension (mm)
L=W	80	c	58.77
X	16	G ₄	52.75

III. ANTENNA PERFORMANCE ANALYSIS

The analysis of the MIMO antenna, including the reflection coefficient, isolation, gain, radiation efficiency, surface current, ECC, CCL, TARC and radiation pattern will be detailed in this section, to highlight the performance characteristics of the design compared with existing literature.

A. Reflection Coefficient

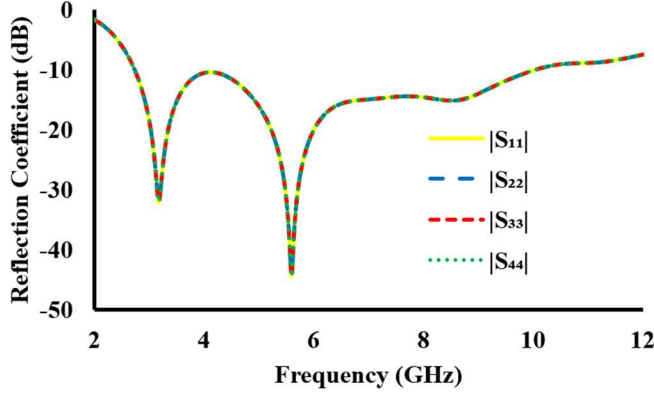


Fig. 4. S-parameters of the MIMO antenna.

Fig. 4 depicts the S-parameters ($|S_{11}|$ to $|S_{44}|$) of the MIMO antenna. Resonant responses occur at frequencies of 3.16 GHz and 5.58 GHz and return losses observed at the resonant locations are -32 dB and -44.4 dB ensures maximum power is radiated. Based on the simulated $|S_{11}|$ curve analysis, it can be deduced that the antenna functions effectively within the frequency range spanning from 2.72 GHz to 10.04 GHz. This range exhibits a bandwidth of 7.32 GHz and all reflection coefficients < -10 dB, confirming good impedance matching, while the overlapping responses indicate that all MIMO antenna elements operate consistently over the intended bands.

B. Isolation and Envelope Correlation Coefficient (ECC)

Fig. 5 (a) depicts the isolation between the elements proposed in this study. The minimum isolation between the antenna elements in the working zone is -19.5 dB and maximum isolation is -32 dB. The examination of isolation between the MIMO antenna ports can also be conducted through the ECC analysis. The ECC quantifies the level of

independence exhibited by the radiation patterns of two antennas. The values of ECC for the antenna shown in Fig. 5 (b). Within the operational region, the suggested antenna exhibits an ECC value below 0.0012, guaranteeing a negligible influence of an individual element on the functionality of other components within a MIMO antenna system.

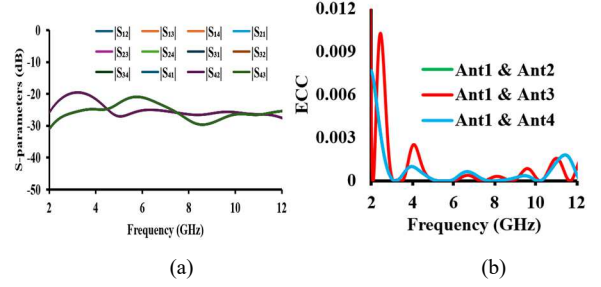


Fig. 5. Analysis of (a) isolation and (b) ECC

C. Gain and Directivity

Fig. 6 illustrates the gain and directivity characteristics of the MIMO antenna across the designated operating frequency range. As depicted in Fig. 6(a) and Fig. 6(b), the antenna achieves a maximum gain of 4.87 dB and a maximum directivity of 5.51 dBi at 5.58 GHz. At the resonant frequencies, the gain and directivity are 3.20 dB and 3.50 dBi at 3.16 GHz, respectively, and 4.87 dB and 5.64 dBi at 5.58 GHz, respectively. The antenna exhibits stable and favorable gain and directivity within the operating band. The corresponding 3D gain and directivity patterns at 3.16 GHz and 5.58 GHz are presented in Fig. 7.

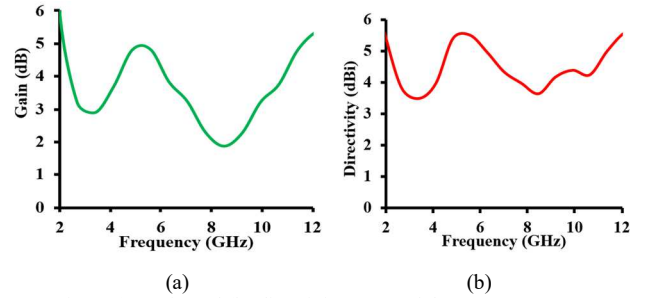


Fig. 6. (a) Gain and (b) directivity curve of the MIMO antenna.

D. Surface Current Distribution

The current distribution of the MIMO antenna is illustrated in Fig. 8 for both the frequencies of 3.16 GHz and 5.58 GHz. The maximum surface current is around 76 A/m at the frequency 3.16 GHz. The maximum value of surface current is around 80 A/m at the frequency 5.58 GHz.

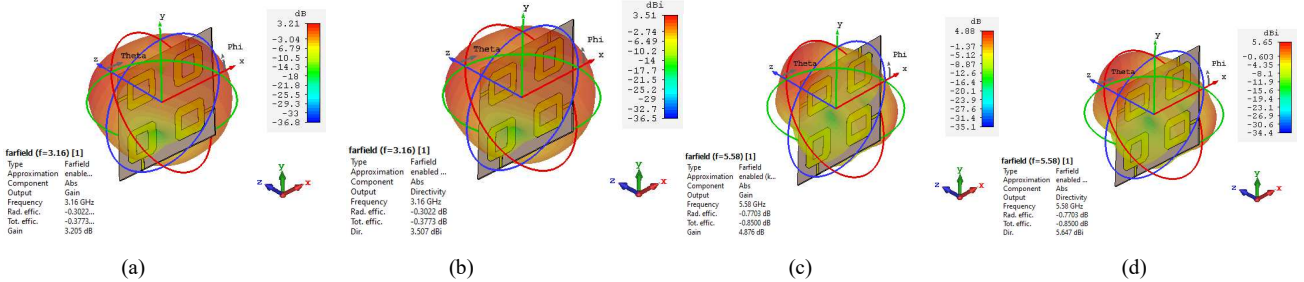


Fig. 7. Analysis of 3D Gain and Directivity at 3.16 GHz and 5.58 GHz.

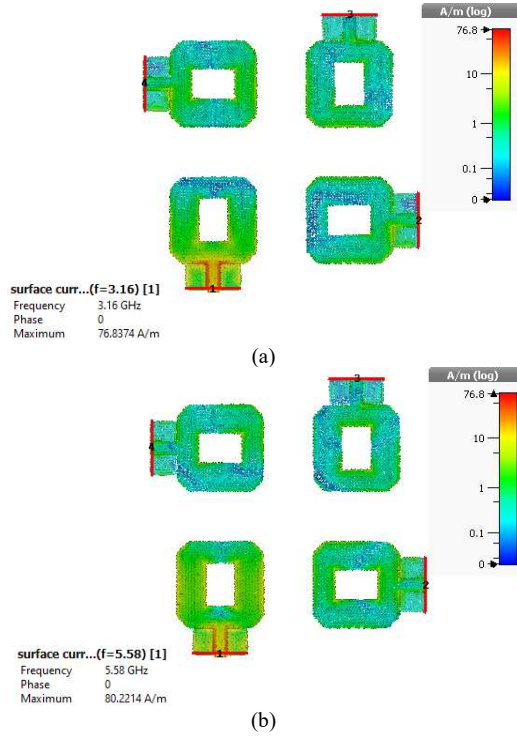


Fig. 8. Current distribution of the MIMO antenna at 3.16 GHz and 5.58 GHz when port-1 is excited.

E. Radiation Efficiency

The measurement of radiation efficiency is a significant factor in characterizing the effectiveness of an antenna in transmitting and receiving RF signals. It is quantified as the proportion of the total power emitted by an antenna to the whole input power received from the source. The MIMO antenna design exhibits a significant radiation efficiency of 76% to 94% in the bandwidth region shown in Fig. 9. So, the antenna can be used suitably within the specified region for which it is designed.

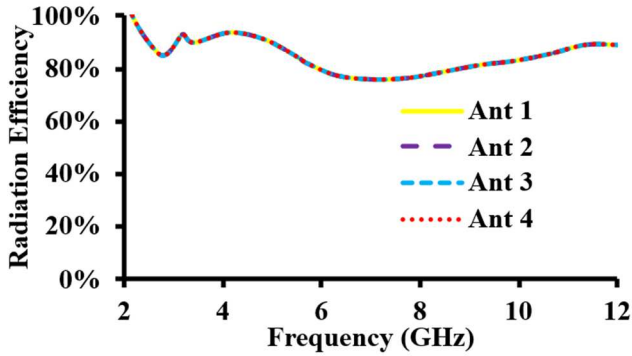


Fig. 9. Efficiency of the four elements MIMO antenna.

F. Diversity Gain (DG)

The data presented in Fig. 10 demonstrates that the simulated DG consistently maintains a proximity of approximately 10 dB throughout the operational frequency range of 2.72 to 10.04 GHz. The MIMO antenna being considered demonstrates advantageous attributes for use in various applications.

G. Electric and Magnetic Field Analysis

As shown in Fig. 11, the four elements are placed orthogonally, with each adjacent element rotated by 90°. The E-field and H-field radiation patterns at 3.16 GHz and 5.58

GHz are presented in Fig. 11(a–d) and Fig. 11(e–h), respectively, using polar plot representations.

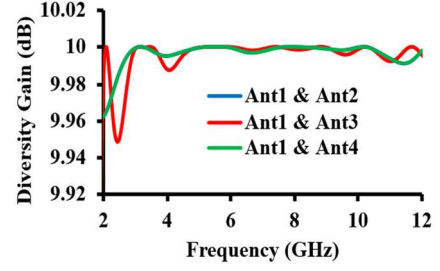


Fig. 10. DG of the four elements MIMO Antenna.

The far-field characteristics shown in Fig. 11(a–d) illustrate the E-field and H-field patterns in the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes at the 3.16 GHz. The E-field and H-field show stable radiation behavior, with the main lobes directed at approximately 139° and 18° in the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes, respectively. The E-field radiation patterns exhibit comparable omnidirectional and lobe-shaped characteristics, with a peak magnitude of around 16.8 dB(V/m) in both planes. The 3-dB angular beamwidth in the $\phi = 90^\circ$ plane is approximately 68.1° , as shown in Fig. 11(a) for port 1 excitation, while the corresponding H-field magnitude is about -35.7 dB(A/m). The corresponding E-field and H-field radiation patterns in Fig. 11(b) show similar E-plane and H-plane characteristics for antenna port 2. Due to the orthogonal configuration of the individual elements, the radiation maxima are oriented in different spatial directions with respect to the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes. The remaining two elements exhibit similar radiation characteristics at each resonant frequency, with corresponding angular rotations in the E-plane and H-plane due to the geometrical symmetry of antenna elements 3 and 4, as shown in Fig. 11(c, d) and Fig. 11(g, h). In the lower-frequency region, no significant side lobes are observed. However, at the higher resonant frequency of 5.58 GHz, side lobes appear, with levels of approximately -3.9 dB and -10.3 dB for the adjacent antenna elements. At 5.58 GHz, as shown in Fig. 11(e), the E-field in the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes demonstrate more directive behavior at port-1. The main-lobe directions are observed at approximately 6° and 21° , respectively, with a peak E-field magnitude of around 19 dB(V/m). The corresponding H-field magnitude is approximately -32.2 dB(A/m) in the $\phi = 0^\circ$ and $\phi = 90^\circ$ planes. The 3-dB beamwidths of 43.6° and 76.7° , shown in Fig. 11(e–h), indicate that the antenna exhibits more directional radiation behavior at the higher resonant frequency.

H. TARC and CCL Analysis

Fig. 12 (a) illustrates the performance evaluation of the MIMO antenna in terms of TARC and CCL over the operating frequency range (2.72–10.04 GHz). When all antenna ports are simultaneously excited the TARC response demonstrates that antenna maintains <-10 dB confirms that mutual coupling and reflection effects are well balanced. Similarly, the CCL plotted in Fig. 12 (b) shows the CCL stays below the acceptable limit of 0.4 bits/s/Hz throughout most of the operating band. This low CCL indicates weak correlation between the radiating elements, which signifies the orthogonal placement of the patch elements.

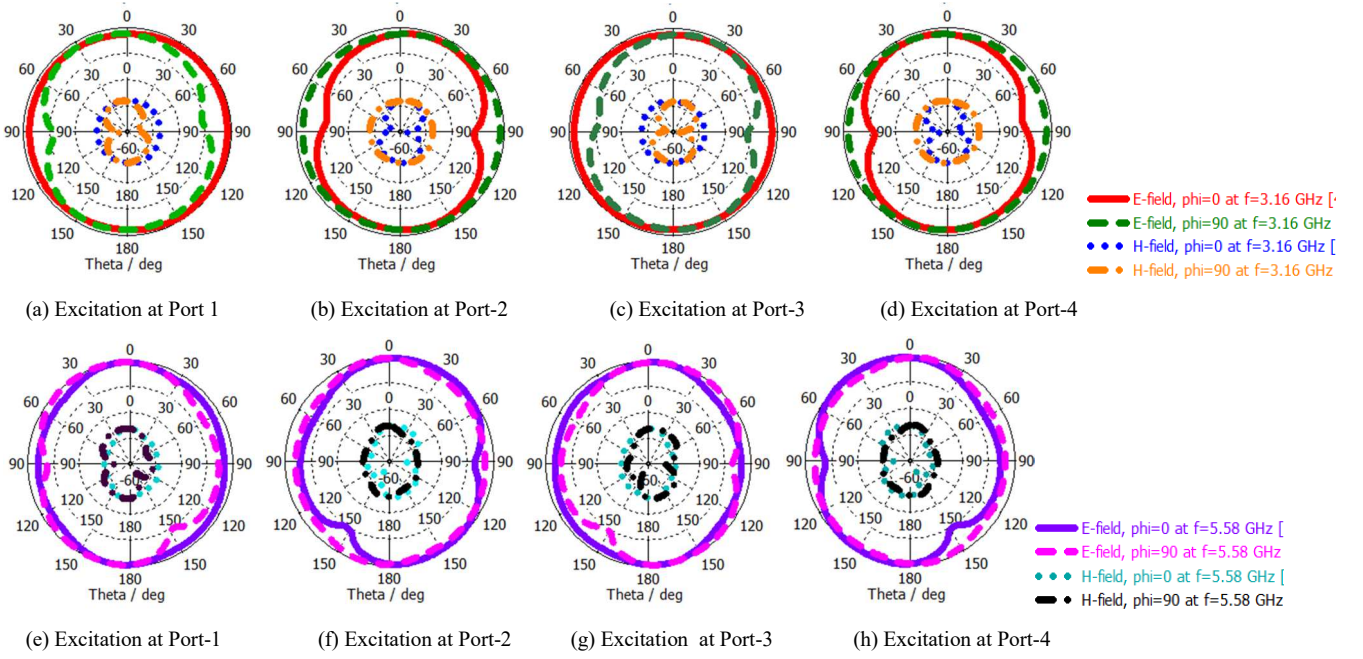


Fig. 11. E-field and H-field analysis of the proposed MIMO antenna at 3.16 GHz and 5.58 GHz for different excited ports.

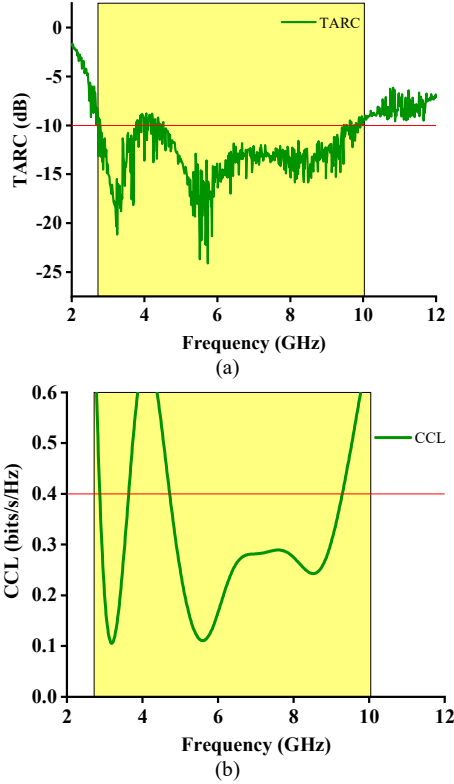


Fig. 12. Analysis of (a) TARC and (b) CCL for the MIMO antenna.

Both findings are derived using S-parameters (S_{ii} , S_{ij}) for N-port MIMO antenna as specified by equation (1-2) [18].

$$TARC = \sqrt{\frac{\sum_{i=1}^N |b_i|^2}{\sum_{i=1}^N |a_i|^2}} \dots \dots \dots (1)$$

where, a_i , b_i =incident and reflected waves at the i^{th} port respectively.

CCL is derived from the correlation matrix R and is given by,

$$CCL = -\log_2(\det(R)) \dots \dots \dots (2)$$

$$\text{where, } R = \begin{bmatrix} 1 - (|S_{ii}|^2 + |S_{ji}|^2) & -(S_{ii}^* S_{ij} + S_{ji}^* S_{jj}) \\ -(S_{ij}^* S_{ii} + S_{jj}^* S_{ji}) & 1 - (|S_{jj}|^2 + |S_{ij}|^2) \end{bmatrix}$$

This analysis confirms effective MIMO performance and suitability for high-capacity wireless applications.

IV. COMPARATIVE ANALYSIS

The proposed design distinguishes itself by achieving a significantly wider bandwidth (7320 MHz) compared to all other listed works in Table III. While many designs consists of large footprints [10]-[16]. Furthermore, the proposed antenna demonstrates excellent isolation of -19.5 dB and an exceptionally low ECC which is better than in [10], [12], [14], [15]. These characteristics, combined with a high efficiency of up to 94%, make the antenna highly suitable for multi-band applications..

V. CONCLUSION

This work presents a high-isolation UWB MIMO antenna designed for Sub-6 GHz, WiMAX, and WLAN systems. The incorporation of a rectangular slot in the patch significantly improves the radiation characteristics of the antenna for the intended communication systems. The MIMO antenna exhibits reflection coefficients of -32 dB at 3.16 GHz and -44.4 dB at 5.58 GHz. It achieves a wide impedance bandwidth of 7.32 GHz, along with strong isolation, ranging from -19.5 dB to -32 dB. The performance of the MIMO antenna was validated using the time-domain solver in CST Microwave Studio. A detailed analysis, supported by relevant existing research, confirms the advantages and feasibility of the design for the intended applications. Therefore, the four-element high-isolation UWB MIMO antenna can be considered a good candidate for the intended applications. Despite its simple structure and compact size of $80 \times 80 \times 0.79 \text{ mm}^3$, the antenna demonstrates excellent gain, directivity, efficiency, and an omnidirectional radiation pattern suitable for wireless systems.

TABLE III. COMPARATIVE ANALYSIS WITH RECENT RELEVANT WORK

Ref.	Size (L×W×h) mm ³	Substrate Material	Operating Frequency (GHz)	BW (MHz)	Reflection Co-efficient (dB)	Centre Frequency (GHz)	Max Gain (dB)	Efficiency (%)	Min. Isolation (dB)	ECC	Applications
[9]	72×72×1.6	FR-4	3.3-6	2700	-42	4.1	> 2.5	> 95	-15	<0.005	5G Sub-6 GHz
[10]	150×75×0.8	FR-4	3.4-3.6	200	-26	3.53	3.9	58-72	-13	<0.1	5G Smart Phones
[11]	120×65×1.6	FR-4	3.3-5	1700	-35	3.8	4.71	NA	-18.8	0.018	5G NR Smartphone
[12]	120×60×1.575	FR-4	0.8-6	5200	-25	0.9, 3.95, 4.75	2.97	> 46	-10	<0.26	Sub-6 GHz 5G
[13]	150×75×1.524	Rogers RT5880	3.38-3.80	420	-25	3.55	3.52	90-92	-15	<0.01	5G Smart Mobile Terminals
[14]	150×72×1.2	FR4- epoxy	4.85-5.26	410	-19.5	5.12	3.7	55	-22	0.68	5G Mobile Phone
[15]	150×80×0.6	FR-4	3.4-3.6	200	-27	3.47	2-4	60	-10	<0.1	5G Smartphone
[16]	150×75×1.6	FR-4	3.21-3.81	600	-23	3.5	3.64	90	-10	0.02	Sub-6 GHz 5G
<i>This Work</i>	80×80×0.79	Rogers RT5880	2.72-10.04	7320	-32, -44.4	3.16, 5.58	4.87	76-94	-19.5	<0.0012	Sub-6 GHz, WiMAX, WLAN

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